



TAIBA N'DIAYE WIND POWER



Document Prepared by AERA GROUP

ON BEHALF OF PARC EOLIEN TAIBA N'DIAYE S.A.U

Project title	Taiba N'Diaye Wind Power
Project ID	2588
Monitoring period	01-August-2024 to 31-July-2025 (both dates included)
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Prepared by	Aera Group

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The main elements of the wind power plant are Vestas V126-3.45 MW turbines, pitch regulated upwind turbines with active yaw and a three-blade rotor. The wind turbine family utilises the OptiTip® concept and a power system based on an induction generator and full-scale converter. With these features, the wind turbine is able to operate the rotor at variable speed and thereby maintain the power output at or near rated power even in high wind speed.

The wind parc is connected to the grid (SENELEC) through a 150m 225KV underground line connected to Tobene substation, which acts as the project substation as well as the grid connection point and with a metering point located at PETN substation.¹

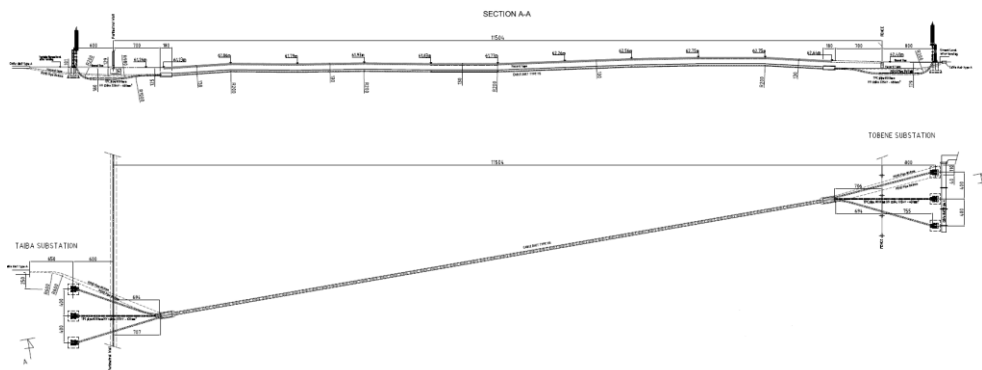


Figure 1: PETN Grid connection

¹ The project doesn't require any transmission line as the Tobene substation acts as the project substation as well as the grid connection point.

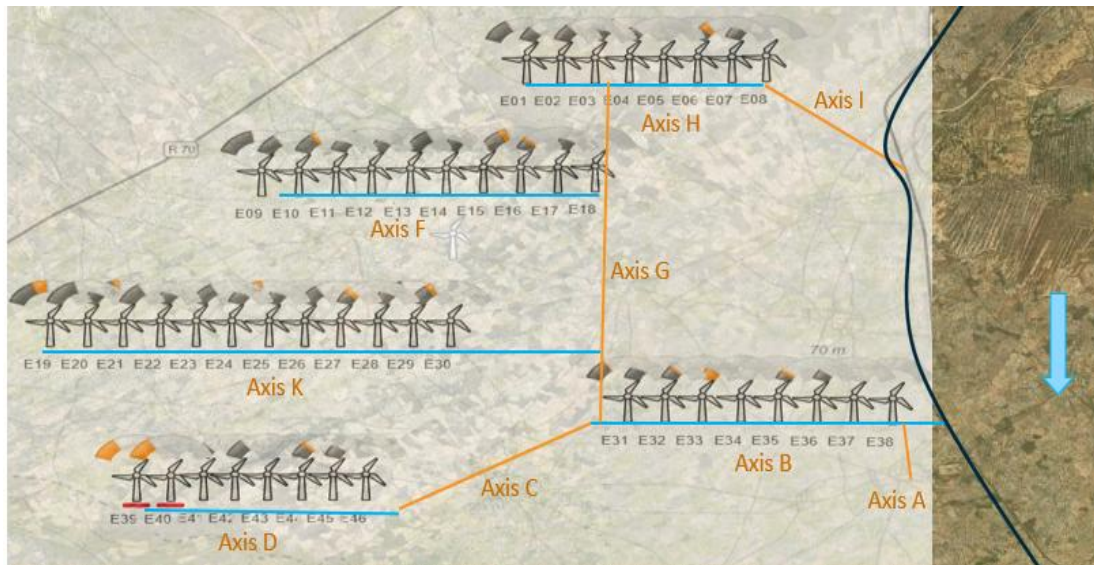


Figure 2: PETN, general arrangement, wind farm

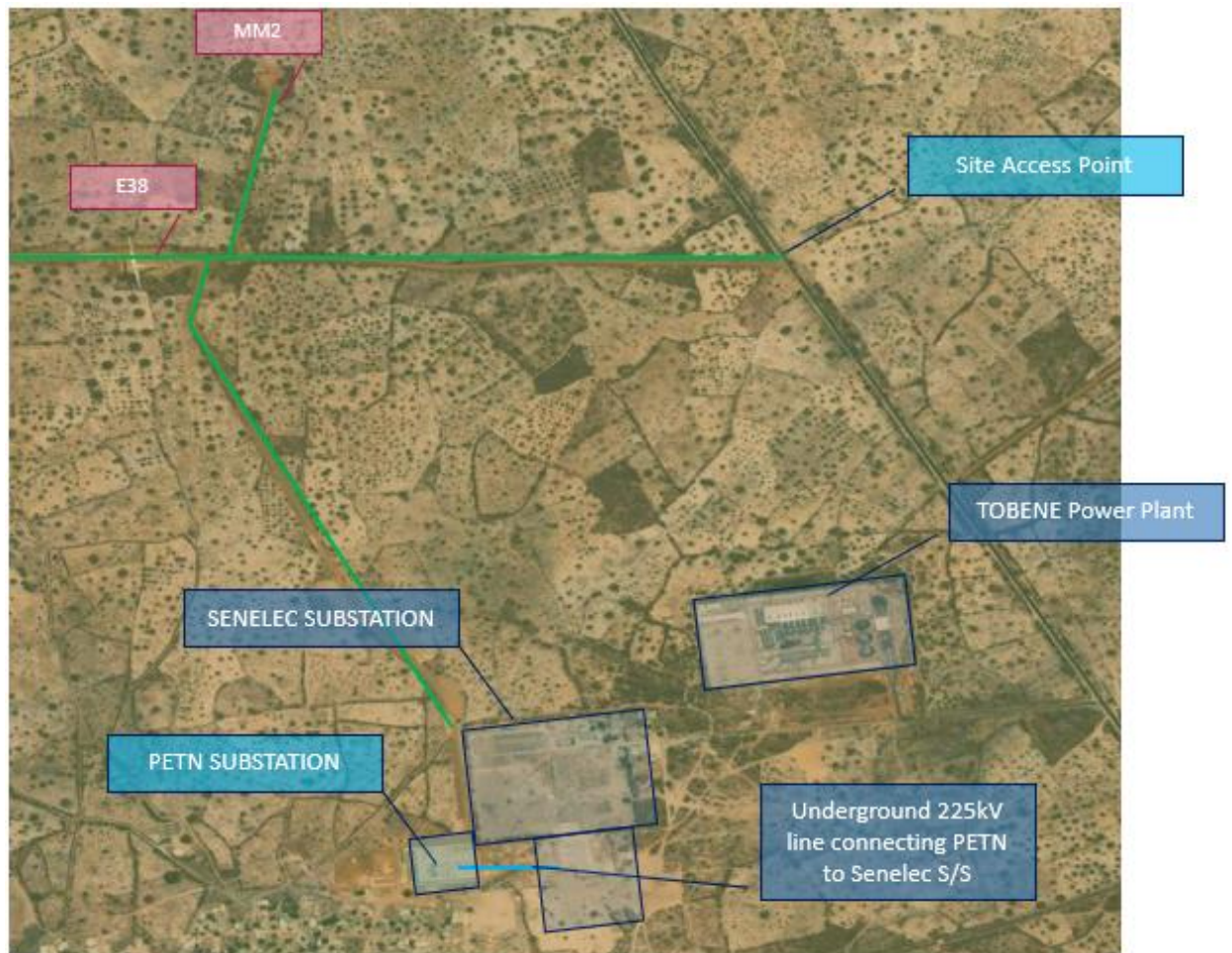


Table 1: Technical specifications of the wind turbines

Rated power	3,450kW
Wind class	IEC IIIA/IEC IIB
Standard operating temperature range	from -20°C* to +45°C with de-rating above 30°C
Rotor diameter	126m
Frequency	50/60 Hz
Converter	full scale

Table 2: Implementation timeline

22-28 December 2012	Public Stakeholders Meeting
21 June 2016	Environmental Impact Clearance
10 October 2016	PPA signature
09 December 2019	Phase 1 Commissioning date
06 November 2020	Phase 2 Commissioning date
30 June 2021	Phase 3 Commissioning date

The total GHG emission reductions generated in this fifth monitoring period is 239,354 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation ²	01-Sep-2013 – 31-Aug-2020	CDM	DET NORSKE VERITAS	Seven years
Validation	09-Dec-2019 – 08-Dec-2029	VCS	4K Earth Sciences Private Limited	Ten years
Verification 1	09-12-2019 – 31-July-2021	VCS	4K Earth Sciences Private Limited	0.6 year
Verification 2	01-Aug-2021 – 31-July-2022	VCS	EARTHOOD SERVICES PRIVATE LIMITED	One year
Verification 3	01-Aug-2022 – 31-July-2023	VCS	EARTHOOD SERVICES PRIVATE LIMITED	One year
Verification 4	01-Aug-2023 – 31-July-2024	VCS	Sustain-Cert	One year

² The project was de-registered on January, 05, 2022

Verification 5	01-Aug-2024 – 31-July-2025	VCS	LGAI Technological Center, S.A. (Applus+ Certification)	One year
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1.3 Sectoral Scope and Project Type

Sectoral scope ³	1
Project activity type	Energy Industries (renewable/non-renewable sources)

1.4 Project Proponent

Organization name	Parc Eolien Taiba N'Diaye S.A.U
Contact person	M. Toumane Mady MBAYE
Title	Finance Director
Address	Immeuble Riviona ,167 Avenue Lamine Gueye, Dakar
Telephone	+221 33 849 73 93
Email	Taiba@lekela.com

1.5 Other Entities Involved in the Project

Organization name	Aera Group
Role in the project	Carbon consultant & offtaker
Contact person	M. Alexandre DUNOD
Title	Head of Certification and Deliveries
Address	28 cours Albert 1er, 75008 Paris, France
Telephone	+33 6 42 96 09 78
Email	a.dunod@aera-group.fr

³ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

Organization name	SENELEC
Role in the Project	Public utility & conceding authority
Contact person	M. Papa Toby GAYE
Title	Head of properties
Address	28 rue Vincens, BP93 Dakar, Senegal
Telephone	+221 33 839 30 30
Email	toby.gaye@senelec.sn

1.6 Project Start Date

Project start date	09-Dec-2019
Justification	The start date of the project activity is the date of commissioning of PETN first tranche. Hence, start of emissions reduction.

1.7 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	09-Dec-2019 to 08-Dec-2029

1.8 Project Location

Country:	Senegal
Region:	Thies
Department:	Tivaouane
Municipality	Taïba Ndiaye

Locations under the influence of the project

Ndomor, Keur Malé, Minam, Mbayéne, Keur Birama, Keur Samba Awa, Keur Mbaye Sénoba. Taïba Mbaye, Same Ndiaye, Baïty Ndiaye, Baïty Gueye, Keur, Madiagne, Taïba Santhie, Keur Assane

The forty-six (46) wind turbines, each occupying 1,400 m², i.e. a total land area of 7 hectares, are served by access tracks that accommodate the high-voltage electrical cables, of a length of approximately 34 km. The project site's geo-coordinates are: Latitude 15°00'36.5"N, Longitude 16°51'32.2"W⁴.

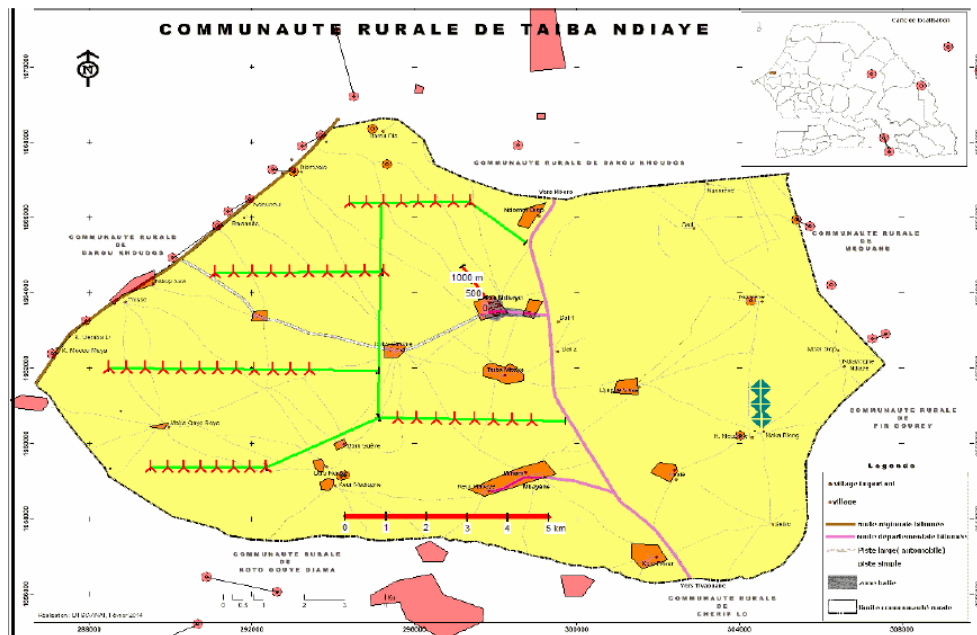


Figure 3: Project site overview

Table 3: Wind turbines coordinates

⁴ <https://www.earthpoint.us/Convert.aspx>

WTG label	WTG type	Hub Height [m]	“Coordinate system, UTM WGS84-28”	
			Easting [m]	Northing [m]
E1	V126-3.45 MW	117	294409	1666417
E2	V126-3.45 MW	117	294835	1666420
E3	V126-3.45 MW	117	295260	1666423
E4	V126-3.45 MW	117	295685	1666426
E5	V126-3.45 MW	117	296110	1666429
E6	V126-3.45 MW	117	296535	1666432
E7	V126-3.45 MW	117	296961	1666435
E8	V126-3.45 MW	117	297386	1666438
E9	V126-3.45 MW	117	291103	1664553
E10	V126-3.45 MW	117	291563	1664556
E11	V126-3.45 MW	117	292023	1664559
E12	V126-3.45 MW	117	292484	1664562
E13	V126-3.45 MW	117	292952	1664565
E14	V126-3.45 MW	117	293404	1664568
E15	V126-3.45 MW	117	293864	1664571
E16	V126-3.45 MW	117	294325	1664574
E17	V126-3.45 MW	117	294785	1664577
E18	V126-3.45 MW	117	295245	1664580
E19	V126-3.45 MW	117	288472	1662016
E20	V126-3.45 MW	117	288922	1662012
E21	V126-3.45 MW	117	289373	1662008
E22	V126-3.45 MW	117	289823	1662004
E23	V126-3.45 MW	117	290273	1662000
E24	V126-3.45 MW	117	290723	1661996
E25	V126-3.45 MW	117	291174	1661992
E26	V126-3.45 MW	117	291624	1661988
E27	V126-3.45 MW	117	292074	1661984
E28	V126-3.45 MW	117	292524	1661980
E29	V126-3.45 MW	117	292974	1661976
E30	V126-3.45 MW	117	293424	1661972
E31	V126-3.45 MW	117	295604	1660681
E32	V126-3.45 MW	117	296075	1660672
E33	V126-3.45 MW	117	296545	1660662
E34	V126-3.45 MW	117	297015	1660653
E35	V126-3.45 MW	117	297485	1660643
E36	V126-3.45 MW	117	297955	1660634
E37	V126-3.45 MW	117	298425	1660624
E38	V126-3.45 MW	117	298895	1660615
E39	V126-3.45 MW	117	289523	1659390
E40	V126-3.45 MW	117	289928	1659390
E41	V126-3.45 MW	117	290333	1659390
E42	V126-3.45 MW	117	290739	1659389
E43	V126-3.45 MW	117	291144	1659389
E44	V126-3.45 MW	117	291549	1659389
E45	V126-3.45 MW	117	291954	1659389
E46	V126-3.45 MW	117	292360	1659389

1.9 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	20.0

TOOL	23	Additionality of first-of-its-kind project activities	03.0
TOOL	07	Tool to calculate the emission factor for an electricity system	7.0
TOOL	ASB0034-2021	Grid emission factor for West African Power Pool	01.0

1.10 Double Counting and Participation under Other GHG Programs

1.10.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.10.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☒ Yes ☐ No

Registered under the CDM UNFCCC, Project ID: 5846⁵

1.11 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.11.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.11.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

⁵ <https://cdm.unfccc.int/Projects/DB/DNV-CUK1330446091.13/view>

1.11.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through the Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

1.12 Sustainable Development Contributions

During this monitoring period, there were no new project activities implemented.

With the construction and operation of this 158.7 MW wind farm, the project ensured access to affordable, sustainable and modern energy for the commune of Taiba N'Diaye (SDG 7) and reduced GHG emissions (SDG 13).

The contribution to these SDGs allows to achieve the pillars of the national sustainable development strategy of Senegal:

- Lettre de Politique de Développement du Secteur de l'Energie
- Stratégie Nationale de Mise en Œuvre de la CCNUCC et Rapport de Contribution Prévue Déterminée au Niveau National (CPDN)

Table 4: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Renewable energy exported to the Senegalese grid from the grid-connected wind farm. (Implemented activities to increase).	417,721 MWh were delivered to the Senegalese grid during this monitoring period	2,106,226 MWh in total were produced and delivered to the grid over project lifetime.
2)	8.8	Promote sustained, inclusive, and sustainable economic growth, full and productive employment and decent work for all	Employments created and labour terms and conditions, such as job-related health and safety and training of employees. (Implemented activities to increase)	5 jobs were created during this monitoring period	32 jobs in total were created during overall project lifetime. All employees are direct with permanent contracts.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	GHG emissions reduction thanks to project (Implemented activities to increase).	The project contributed to Emission Reductions amounting to 239,354 tCO2e for this monitoring period	Total emission reductions for this project amount to 1,206,865 tCO2e to date.

1.13 Commercially Sensitive Information

There is no commercially sensitive information that has been excluded from the public version of the monitoring report.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

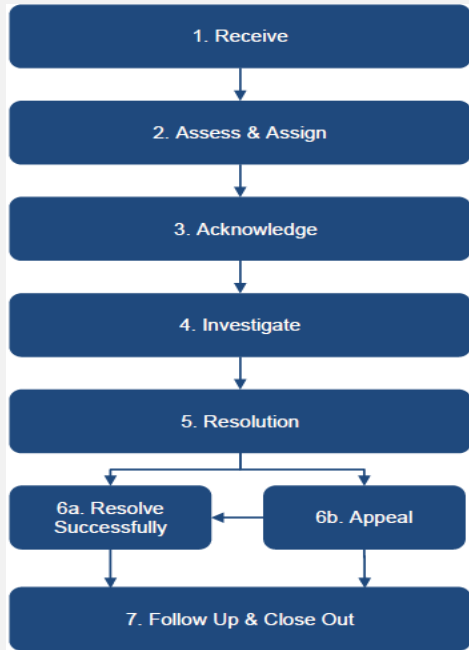
Stakeholder Identification	The choice of each category of stakeholders is justified by its central position in the process engaged; in particular the role it is called upon to play, its intermediary position or even its more or less direct impact through the implementation of the project. Since validation, the stakeholder make up has not changed, thus this is not applicable.
Legal or customary tenure/access rights	As per livelihood restoration plan, land is secured through long-term lease agreements (baux emphytéotiques) with the State of Senegal. Additional land owned by the national energy company, SENELEC were required for the Project substation and energy delivery facilities. Rights to these lands were secured as per SENELEC lease agreement. PP and impacted population agreed on financial compensation following the loss of economic assets. All compensation has been made before project launch⁶.
Stakeholder diversity and changes over time	As per local stakeholders' consultation report, different institutions and social groups were invited to the stakeholder consultation meeting. Since the LCS report there was no change in the make-up of the stakeholders' groups. Thus, this is not applicable.
Expected changes in well-being	The project has several benefits for the region such as job creation, renewable energy generation, supporting local infrastructures (the construction of a school, skips for waste management, road improvement).

⁶ To take charge and apply national regulations on resettlement, the expenses assessment commission was mobilized under the direction of the Prefect of Tivaouane.

Location of stakeholders	All stakeholders are located in the region of Thiès, Senegal.
Location of resources	Impacted lands are located in Taiba Ndiaye, where the power plant has been built.

2.1.2 Stakeholder Consultation and Ongoing Communication

Ongoing consultation	By communicating all contact details of the project proponent and ESIA consultant during the meetings, stakeholders were informed about the creation of the grievance mechanism through which any complaints, suggestions, recommendations etc. could be communicated. During this monitoring period, one complaint was received.
Date(s) of stakeholder consultation	From 22-December-2012 to 28-December-2012
Communication of monitored results	During this monitoring period, there was 1 complaint received. As per Grievance register report : In October 2024, during the development of storage areas for the power plant materials, the workers in charge of the work deposited sand dunes in someone's fields. The person requested PETN to take the necessary steps for an emergency correction. On the site, the observation made by the CLO is that the incident is not recent and dates from the construction (around 2019). Moreover, weeds from past winters have grown on the topsoil deposit. The first question that comes to mind is why the complainant has not filed a grievance since the impact occurred, knowing that dozens of cases of this nature have been recorded, processed and closed. The following measures were adopted: • Mobilize a machine to level the topsoil. • Relocate the fence so that the complainant can recover the entire area lost. These measures were accepted by the complainant and were implemented on September 11, 2025. For 2025, there was no complaint received. All grievances have been closed to date.

Consultation records	A formalized and official PETN Grievance Mechanism Procedure (GMP) is established in order to manage grievances that may originate from the various project related activities. The GMP will address grievances that originate from the wider public (community) relating to any aspect of the project.  <pre> graph TD 1[1. Receive] --> 2[2. Assess & Assign] 2 --> 3[3. Acknowledge] 3 --> 4[4. Investigate] 4 --> 5[5. Resolution] 5 --> 6a[6a. Resolve Successfully] 5 --> 6b[6b. Appeal] 6a --> 7[7. Follow Up & Close Out] 6b --> 7 </pre> Grievances can be filed through any of the following channels: • Via verbal communication to any PETN or affiliated site employee - Employees will be advised to direct individual grievances to a site representative designated to receive such input. Non-employees can speak to any site manager to discuss grievance. If needed, individuals will be assisted by PETN representatives in the applicable local language. • Via Telephone to a dedicated hotline (+221 33 849 73 93) • Via e-mail : Taiba@lekela.com
Stakeholder input	As there was no complaint about the design of the project, there was thus no need to modify the power plant.

2.1.3 Free, Prior, and Informed Consent

Consent	Both PP and impacted people agreed on financial compensation. Beyond this financial compensation, a reintegration and assistance policy for impacted people could be implemented to promote the sustainability of resettlement and beneficial socio-economic effects.
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	There are no ongoing or unresolved conflicts. As explained in section 2.1.2, there was 1 grievance during this monitoring period, and it was successfully resolved.
Outcome of FPIC	The project received all necessary environmental clearance before project launch. The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

Grievances received	Resolution and outcome
During this monitoring period, there was 1 complaint received	During the development of storage areas for the power plant materials, the workers in charge of the work deposited sand dunes in someone's fields. The person requested PETN to take the necessary steps for an emergency correction. The incident is not recent and dates from the construction (around 2019). Following the audit, a working session was held with the complainant at the level of his field. After discussion with impacted stakeholder, the following measures were adopted: • Mobilize a machine to level the topsoil. • Relocate the fence so that the complainant can recover the entire area lost. These measures were accepted by the complainant and were implemented on September 11, 2025. Grievances and/or comments can easily be filed through any of the following channels: • Via verbal communication to any PETN or affiliated site employee - Employees will be advised to direct individual grievances to a site representative designated to receive such input. Non-employees can speak to any site manager to discuss grievance. If needed, individuals will be assisted by PETN representatives in the applicable local language. • Via Telephone to a dedicated hotline (+221 33 849 73 93) • Via e-mail : Taiba@lekela.com

2.1.5 Public Comments

There was no public comment received from stakeholders outside of the 30-day public comment period. Thus, this is not applicable.

Summary of comments received	Actions taken
Not applicable.	Not applicable.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Lekela is a subsidiary of Infinity power, which is one of the largest African renewable energies' companies operating in Senegal, Egypt & South Africa. The company combines the expertise and experience of its founding shareholder companies, Masdar and Infinity, specializing in power generation projects through renewable energy sources on utility-scale, commercial and industrial scales.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	All compensation was made, and in addition, PP helps farmers to strengthen their production capacity by helping to set up market gardening areas as well as through the establishment of nurseries, in collaboration with the Water and Forestry services and the women's groups, to replace the trees felled in order to correct the imbalance, especially with fruit trees.
Risks to stakeholder participation	Not applicable	As per ESMP, there is no risk related to stakeholder participation.
Working conditions	Not applicable	As per E&S report of Q2 2025, Regular inspections of employees' working conditions have been conducted. These inspections make it possible to identify all non-compliances in the working conditions of employees and to take corrective measures. No issue was detected. Furthermore, the E&S report shows that there is no workers grievance. Several safety trainings have also been provided to workers from January to July 2025. All employees have necessary equipment.

Safety of women and girls	Not applicable	As per grievance register as well as on site audits, there has never been any prejudice towards women and girls.
Safety of minority and marginalized groups, including children	Not applicable	As per ESMP, there is no minority nor marginalized groups.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Not applicable	As per ESMP, there is no release of pollutants during the operational phase of the power plant.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁷	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	As per latest environmental and social report, there is no issue related to discrimination. This is also verified during audits. This is also clearly prohibited by the company's code of conduct.
Sexual harassment	No risk identified	As per company's code of conduct, it is clearly mentioned that PP "supports the rights of all people to seek, obtain and hold employment without harassment and all Staff therefore have a responsibility to behave in a manner which is not, nor is likely to be perceived as, offensive to others", and it is PP's "policy to make every effort to provide a working environment free from harassment and intimidation (whether sexual, racial or otherwise)."
Gender equity in	No risk identified	Both women and men are employed by the company.

⁷ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

labor and work		
Forced labor	No risk identified	As per PP's code of conduct <i>"maintains a zero-tolerance policy towards child, illegal, abusive, or forced labour and human trafficking across our operations. We are fully committed to upholding all applicable employment laws, ensuring fair and competitive wages, and conducting our business in a way that respects and promotes individual human rights. This commitment is guided by our internal policies as well as internationally recognized frameworks, including the United Nations Declaration of Human Rights and the core conventions of the International Labour Organisation (ILO)"</i> .
Child labor	No risk identified	Verified during audits. Also prohibited by PP as per Code of conduct.
Human trafficking	No risk identified	Verified during audits. Also prohibited by PP as per its code of conduct.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	There are no identified indigenous peoples at power plant's site. Furthermore, Lekela is operating in Senegal, which is an ILO state member and which signed 10 of 10 Fundamental Conventions, and 3 of 4 Governance Conventions (PRORORITY) ⁸

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	There are no indigenous people living at project's site.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	All financial compensation was made to impacted stakeholders before project launch.

⁸ https://normlex.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103013

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	Not applicable as there no benefit sharing plan. As per long-term lease agreements (baux emphytéotiques) with the State of Senegal. Furthermore all PAPs (Person affected by the project) were compensated, as per payments document. Indeed, both PP and Impacted stakeholders agreed on financial compensation. Furthermore, PP helps revegetate the site by strengthening the means of agricultural production, particularly through fruit tree cultivation.
Benefit sharing during the monitoring period	Not applicable as per lease agreement.

2.4 Ecosystem Health

	Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
Impacts on biodiversity and ecosystems	- Direct mortality from collision with wind turbine blades; - Disturbances, which result in a “barrier effect”, a distance or even sometimes in critical situations a loss of habitats.	• Choice of site: this is the main factor that allows reducing or eliminating the majority of impacts on natural environments and therefore on birds; • Positioning of wind turbines to avoid barrier or funnel effects. An orientation of the parks parallel to the migratory axes effectively reduces the negative effects on migratory birds. • Choice of the period of works and planning of the site according to the species calendar, in particular outside the breeding periods of the most sensitive local species, i.e. between April and September inclusive. A new Post Construction Fatality Monitoring (PCFM) protocol that aimed at raising the standard and extent of fatality monitoring is fully operational PCFM. 17 carcasses found, during Q2 2025 including 03 bats. No endangered species.

Soil degradation and soil erosion	No risk identified	As per ESMP, during the operational phase, there is a low risk of soil and subsoil pollution: presence of oils in wind turbines (approximately 1500 l/wind turbine), oils in transformers. Retention methods adapted to the level of the installations concerned limit this risk.
Water consumption and stress	No risk identified	As per ESMP, there is no risk related to water consumption and stress.

2.4.1 Rare, Threatened, and Endangered species

Species or habitat	Not applicable as there is no rare, threatened and endangered species as per ESMP.
Areas needed for habitat connectivity	N/A

2.4.2 Introduction of species

There were no new species introduced. All trees species planted in the project area are the same as those that were there before project implementation. Indeed, the project is located in an area where *Mango trees*, *peanut fields* are planted. Thus, this is not applicable.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A	N/A	N/A	N/A

2.4.3 Ecosystem conversion

The project is not an ARR, ALM, WRC or ACoGS thus, this is not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	Not applicable as the project is not an ARR, ALM, WRC or ACoGS

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

During this monitoring period, there was no major breakdown which could have impacted production and/or the significant change in emission reductions. This is verified through the monthly energy generation data (invoices), which are consistent from one month to another as well as maintenance reports (provided to VVB).

Furthermore, there was no change in project proponent or other entities involved.

In total, during this monitoring period, the project generated 417 GWh of renewable energy, which led to the reduction of 239,354 tCO₂e.

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation applied during actual or previous monitoring periods.

3.2.2 Project Description Deviations

There is no project description deviation during actual or previous monitoring periods.

3.3 Grouped Projects

Not applicable as the project is not a grouped project.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes

☒ No

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”
Source of data	As per ASB0034-2021
Value applied	0.573
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” version 07.0
Purpose of data	Calculation of baseline emissions
Comments	-

4.2 Data and Parameters Monitored

Data / Parameter	EG _{facility,y}			
Data unit	MWh/yr			
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y			
Source of data	Measured directly with electricity meter(s) at substation.			
Description of measurement methods and procedures to be applied	Electricity outputs will be electronically stored and invoiced monthly to Senelec by PETN, after reconciliation with internal meter which readings are recorded on a record sheet by the Technical/Engineering/Maintenance Department under the Plant Manager's authority			
Frequency of monitoring/recording	Continuous measurement and at least monthly recording			
Value monitored	Year	2024 (from 01/08/2024)	2025 (until 31/07/2025)	Total
	EG _{facility,y}	118,569	299,152	417,721
Monitoring equipment	Bi-directional electronic electricity meters measuring the net electrical energy delivered to Senelec with the following specifications:			

	• Make: Itron • Model: SL7000 • Accuracy class: 0.2S ($\pm 0.2\%$) • Serial ID: 84406427 (main) - 84406426 (back-up)
QA/QC procedures to be applied	Cross check of measurement results with invoiced electricity. Calibration of the meters will be carried out after each deviation of more than $\pm 0.5\%$ between the 2 meters as per manufacturer & PPA specifications.
Purpose of the data	Calculation of baseline emissions
Calculation method	Electronic recording
Comments	Previous EnergiExpress inspection/testing was undertaken on 20/12/2023 and valid until 19/12/2024. Latest inspection/testing was done on 16/12/2024 and valid until 15/12/2025, as per report.

4.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources, version 20.0 whereby it is stated that:

“All data collected as part of monitoring should be archived electronically and be kept at least for 2 years after the end of the last crediting period”.

Therefore, the quantity of net electricity generation supplied by the project plant to the grid will be reliably monitored through calibrated electricity meters and cross-checked with sales records as per internal monitoring practices.

Internal auditing

Internal auditing procedures are conducted by a dedicated team in charge of recording and reporting of the relevant data. All the data regarding energy production and consumption is then consolidated through signed energy reading sheets. During this monitoring period, there was no non-conformities detected, as per calibration report.

Monitoring team and training

Data collection, consolidation and results analysis will be undertaken by a dedicated team adequately trained, well aware of VCS requirements. This team will not have any hierarchical relationships or dependence links with all entities involved to measure net electricity supplied to the grid and to assure the correct operation and maintenance of the measuring equipment. This independence shall guarantee the integrity of the work that will be done.

Quality management procedures

The implementation of a system of supervision and control that will transfer all real-time monitoring information for off-site recording will allow the operator to access and retrieve safely all Project data regardless of eventual incidents.

Meter measurements by Senelec main/back-up shall not exceed $\pm 0.5\%$ difference, beyond which dysfunctional meter will in all cases need to be identified by Senelec and PETN, adjusted or replaced within 48 hours in accordance with manufacturer guidelines. After each deviation of more than $\pm 0.5\%$, a test and calibration of the meters will be carried out for each meter, certified by a third party.

Non-conformance data during uncalibrated periods will systematically be discounted by the corresponding error margin.

Table 5: Characteristics of the meters:

Make	Itron
Model	SL7000
Quantity	2
#ID1	84406427
#ID2	84406426
Accuracy	0.2s
Calibration dates	16/12/2024 valid until 15/12/2025. Previous calibration was undertaken on 20/12/2023 and valid until 19/12/2024
Error identified for main meter during last calibration	-0.02%
Error identified for back-up meter during last calibration	-0.02%
Calibration certificate number	0242-EE & 0243-EE (current) 0148-EE & 0149-EE (previous)

Figure 4: Line Diagram



5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The monitoring period for which GHG emission reductions were achieved spans 01/08/2024 to 31/07/2025. Baseline emissions are firstly calculated based on the difference between the total quantities imported and the total quantities produced that were recorded by the main meter and the back-up meter and which are extracted from the monthly reconciliations, once total production and consumption is available for each month, baseline emissions are calculated based on §4.1 methodological approach as the product of (i) the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y ($EG_{facility,y} = \text{Exports} - \text{Imports}$, in MWh/yr) and (ii) the combined margin CO₂ emission factor for grid connected power generation in year y ($EF_{grid,CM,y}$):

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO ₂ /MWh)

$EG_{PJ,y} = EG_{facility,y}$ i.e. the quantity of net electricity generation supplied by the project plant to the grid, as monitored and displayed below on a monthly basis.

$EF_{grid,CM,y} = 0.573$ tCO₂/MWh, as the Project Proponent has applied approved standardized baseline- ASB0034-2021 Grid emission factor for West African Power v1.0.

Year	BE_y	$EF_{grid,CM,y}$	$EG_{facility,y}$
	t CO ₂	tCO ₂ /MWh	MWh
2023 (from 01/08/2024 until 31/12/2024)	67,940	0.573	118,569
2024 (from 01/01/2025 until /31/07/2025)	171,414	0.573	299,152
Total	239,354		417,721

5.2 Project Emissions

No project emissions are considered as the project activity only involves renewable electricity generation from the wind power plant. Hence according to ACM0002 guidelines $PE_y = 0$.

5.3 Leakage Emissions

As stated in the applicable methodology, no leakage emissions are considered, therefore $LE_y = 0$.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

As per ACM0002 methodology, v20.0 paragraph 54, Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \quad \text{Equation (17)}$$

Where:

ER_y = Emission reductions in year y (t CO₂e/yr)

BE_y = Baseline emissions in year y (t CO₂/yr)

PE_y = Project emissions in year y (t CO₂e/yr)

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
01-August-2024 – 31-December-2024	67,940	0	0	67,940	0	67,940
01-January-2025 – 31-July-2025	171,414	0	0	171,414	0	171,414
Total	239,354	0	0	239,354	0	239,354

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
01-August-2024 – 31-December-2024	107,390	67,940	-37%	Due to maintenance activities plus difference between theoretical calculation and actual data.

01-January-2025 – 31-July-2025	150,346	171,414	-14%	Due to maintenance activities plus difference between theoretical calculation and actual data.
Total	257,735	239,354	-7%	Due to difference between theoretical calculation and actual data.

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

N/A

APPENDIX 2: LIST OF EMPLOYEES

PETN STAFF LIST

Number	Employee #	Gender	Contract	Position	Entry date
1	SEN103	Female	Permanent	Office Manager/Executive Assistant	01/09/2018
2	SEN104	Male	Permanent	Community Liaison Officer	01/09/2018
3	SEN106	Female	Permanent	Accountant	10/12/2018
4	SEN110	Male	Permanent	ESG Manager	04/02/2019
5	SEN112	Female	Permanent	Office Assistant	20/05/2019
6	SEN113	Male	Permanent	Driver	20/05/2019
7	SEN114	Male	Permanent	Financial Controller	07/10/2019
8	SEN116	Male	Permanent	Maintenance Planner	06/01/2020
9	SEN117	Male	Permanent	Substation Operator	10/02/2020
10	SEN118	Male	Permanent	Electrician	26/02/2020
11	SEN120	Male	Permanent	HSE Manager	13/04/2020
12	SEN121	Female	Permanent	HC Manager	01/02/2021
13	SEN122	Male	Permanent	General Manager	06/09/2021
14	SEN123	Male	Permanent	Driver	01/10/2021
15	SEN124	Male	Permanent	Substation Operator	06/04/2022
16	SEN125	Male	Permanent	Substation Operator	06/04/2022
17	SEN127	Male	Permanent	Procurement Specialist	02/05/2022
18	SEN128	Female	Permanent	Junior accountant	02/01/2024
19	SEN129	Male	Fixed-term	Driver	10/09/2024
20	SEN130	Male	Permanent	BD Manager	01/01/2025
21	SEN131	Male	Permanent	Electrician	01/04/2025
22	SEN132	Male	Permanent	Finance Director	10/06/2025
23	SEN133	Male	Permanent	Plant Manager	07/07/2025